TEZPUR UNIVERSITY

Spring Semester End Examination, 2022

Design and Analysis of Algorithms (CO 206)

Full Marks : 60 Duration : 3 hours

You may answer as many questions as you like but maximum you get is 60.

1. Following is a variation of Knapsack problem. [7]

   Input: n items of non-negative weights $w_1, w_2, \cdots, w_n$, profits $p_1, p_2, \cdots, p_n$, two positive integers C and S.

   Output (yes/no:) Does there exist a subset of items with total weight at most C that correspond to a profit of S?

   Does the above problem belong to the class NP?

2. Prove that for any amount $x \geq 34$ rupees, you can give a change of x using only 5 and 7 rupees coins. Write an efficient algorithm to find how many 5 and 7 rupees coins are required to give an exchange of x rupees? For example, we need 4 coins of 7 rupees and 3 coins of 5 rupees for a change of 43. [10+10]

3. Describe the running time of the following pseudocode in Θ notation in terms of n [7]

```
for (i = n; i ≥ 1; i = i/2)
    for (j = 0; j ≤ i; j = j + 1)
        sum=sum+j;
```

4. Answer True / False and provide a counterexample or a short justification. [8×4]

   a) Consider an undirected graph $G(V, E)$ with s as the source vertex, without any negative cycle. All the edge weights are positive except those adjacent to s which may be negatively weighted. Dijkstra's algorithm will generate the correct output for G.

   b) If every CUT of a directed graph G has a single crossing edge, then the single source shortest path problem on G can be solved in $O(|v|)$ time.

   c) In the Knapsack problem with k items, we decrease the weight of the objects by 1 and decrease the capacity by k. The total profit will be more than the profit in the original situation.

   d) If Π_1 is NP-complete and $\Pi_1 <_p \Pi_2$, then Π_2 is NP-complete.

   e) In the matrix chain multiplication problem, if the dimension of the matrices are in non-decreasing order, then multiplying the matrices sequentially from right to left will be the most efficient way to get the final product.

   f) If $f(i) = \max\{f(i-1) + 2f(i-2), 3f(i-3)\}$ and $f(1) = 1, f(2) = 1$ and $f(3) = 2$, then $f(n)$ can be computed in polynomial time.

   g) If all the code words in a prefix-free code are reversed, the resultant code is still prefix-free.

h) If edge weights of a connected weighted graph are not all distinct, the graph must have more than one minimum spanning tree.

5. Suppose you are given the filled up table generated by Matrix chain multiplication algorithm for a given sequence of matrices $M_1 \times M_2 \times \cdots \times M_n$. You are not given the dimension of the matrices. You are asked to determine the order in which the matrices are to be multiplied. Is it possible? Why or why not? [10]

6. Draw two graphs G_1 and G_2(each with 6 vertices and 12 edges) for the following two cases: [5+5]

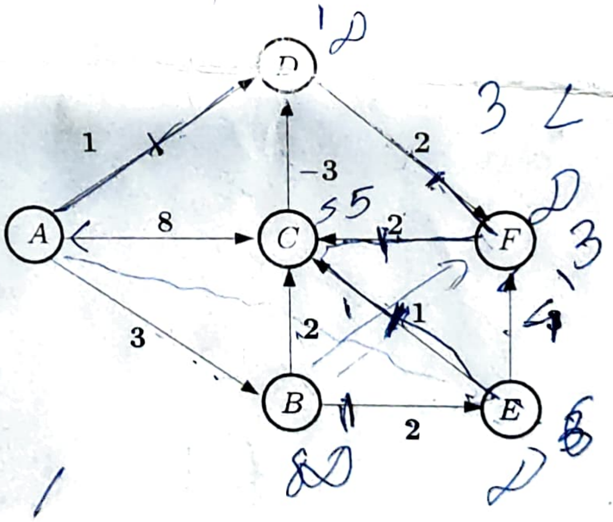
   a) Kruskal's algorithm calls Find(u,v) 7 times to build the MST of G_1.

   b) Kruskal's algorithm calls Find(u,v) 10 times to build the MST of G_2.

7. Solve the following instances of the Knapsack problem using dynamic programming technique for a Knapsack capacity 15. Show the table and the bottom up algorithm only. [10]

Item	1	2	3	4	5
Value	5	8	10	6	4
Weight	2	3	4	1	2

8. Apply Bellman-Ford's algorithm for the following graph to find the distances of all the vertices from A. What will be the difference if Dijkstra's Algorithm is applied on the same graph? [10+10]



A B C D E F G

A,B | B,A
1 | 2